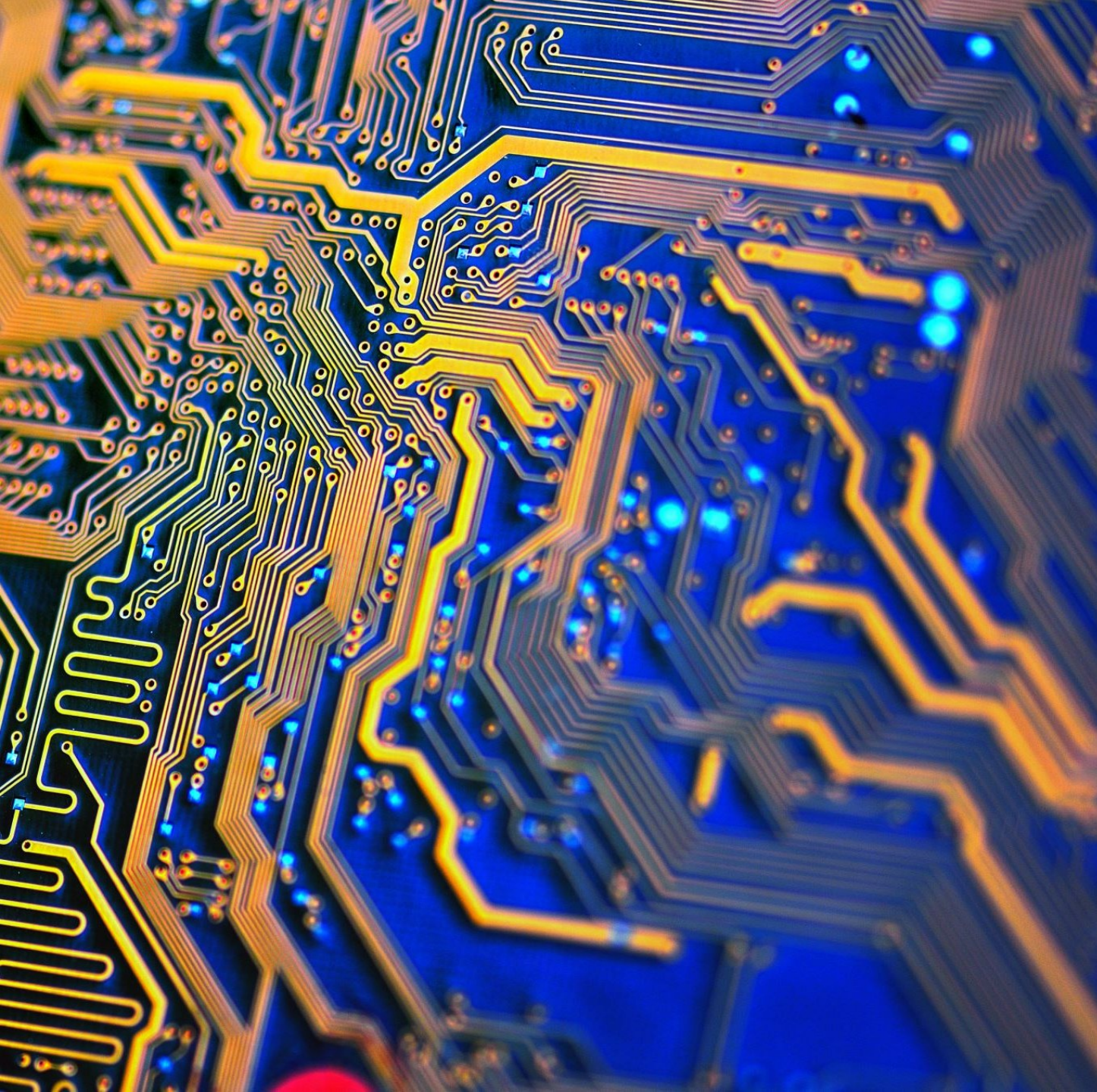




Data science project



ANDY TAI



Welcome to the MSc Data Science Project

- Core component of your MSc (worth 60 credits)
- Major independent research and practical investigation
- Integrates knowledge, skills, and methods
- Output: Final Project Report plus viva exam
- Challenging but rewarding: your 'showpiece' for employers



Importance of the Project

- Apply advanced data science methods
- Develop independence, organisation, and critical thinking
- Showcase ability to plan and execute complex work
- Involves coding, data analysis, and evaluation
- Prepares you for research, industry, or doctoral study



Module Aims

- Design and conduct a substantial research project
- Demonstrate advanced knowledge in specialised data science area
- Develop research skills: planning, writing, presenting
- Show ability to debate ideas in viva-style questioning
- Strengthen understanding of ethics and good practice



Skills You Will Develop

- Critical literature review and synthesis
- Data collection, preparation, and management
- Application of modelling and evaluation methods
- Professional report writing and technical communication
- Academic integrity and independent learning



Requirements to Pass

- Must achieve $\geq 50\%$ overall grade
- Must pass the viva (no viva = fail)
- Assessment split across 4 components (10%, 10%, 70%, 10%)
- Practical coding evidence required
- Strong attendance and engagement expected



Deadlines, Late Submissions, Extensions

- Late work: -10 points/day (up to 5 days) or till its 50%
- BUT ... After 5 days = 0% grade
- Short-term extensions: up to 7 days (not ethics quiz)
- Apply before due date ~~via SASH form~~
- Exceptional Circumstances possible with evidence



If You Fail the Module

- Referral (re-submission) capped at 50%
- If referral fails → repeat module, with cost
- Deferral possible if EC approved (counts as first attempt)
- Serious failures may delay degree award
- Always communicate early with supervisor



Where to Get Support

- Supervisor/Module Leader:
 - main academic and personal tutor during project
 - general queries and oversight

Choosing Your Project

- Complete Choosing a Project form early
- Project must involve analysis of data
- Should address a clear research question or hypothesis
- Select a topic you will enjoy (~600 hours of work)
- Discuss and agree with supervisor before starting



Possible Project Types

- Analyse a new dataset from a real-world context
- Use test dataset to develop a model
- Compare multiple models or approaches
- Develop new statistical or mathematical techniques
- Work with company datasets or industry applications

What makes a good project? Make sure you do the following...

Your project must include the following work:

1. **Practical Work** – this requires you to do some coding
2. **Literature Review**
 - undertake a literature search
 - critical analysis of the literature and relate it to your project
 - references
3. **Evaluation and Interpretation**
 - Analyse and draw conclusions from your work
 - Use domain knowledge about your specific application

Characteristics of a Good Project

- Clearly defined research question
- Appropriate, high-quality dataset – **understand your data**
- Strong pre-processing and methodological choices
- Suitable evaluation metrics and critical conclusions
 - **Use suitable models** for the task and dataset
 - Have an appropriate training routine that takes into account overfitting/underfitting
 - Produce **appropriate metrics**
- Evaluation and **in-depth analysis**
- Evidence of optimisation and exploration of models – **Explore improvements (optimise each model/method you use)**

What makes a good project? (SUMMARY)

Data

- ▶ Data selection (creation) appropriate for the task
- ▶ Data cleaning
- ▶ Pre-processing
- ▶ Encoding/embedding

Models

- ▶ Appropriate for the task/data
- ▶ Why have you chosen the models?
- ▶ What was used in the literature for this application?

Model modification

- ▶ Explore model improvements
- ▶ Explore Hyperparameters
- ▶ Relate them to what is done in the literature

Training

- ▶ Model dependent
- ▶ Parameter tuning (needs to be documented)
- ▶ Overfitting/underfitting checks
- ▶ Training/testing/validation

Evaluation

- ▶ Appropriate metrics chosen (more than one)
- ▶ Rational for chosen metrics
- ▶ Domain dependent
- ▶ Visualisation



Choosing a Dataset

- Must be suitable for research task and ethics-approved
- Supervisor must agree dataset choice
- Sources: World Bank, Our World in Data, Kaggle, company datasets
- Consider sample size, attributes, ethical issues
- Dataset quality is critical for success



Ethical Requirements

- Must apply for UH Ethics approval if involving human data
- Approval needed before surveys, interviews, or user testing
- Responsibility lies with you to verify dataset source
- Personal data must be anonymised with permission
- Non-compliance can lead to failure/misconduct



Applying for Ethics Approval

- Complete UH forms well in advance (min. 3 weeks)
- Supervisor checks documents before submission
- Provide detailed purpose, methods, and sample questions
- Ensure anonymity and informed consent
- Approval delays can risk your timeline



Health & Safety Considerations

- Comply with UH H&S regulations
- Risks in data collection, coding, or fieldwork
- Awareness of digital safety (GDPR, data security)
- Must declare potential hazards in planning
- Ignoring safety can invalidate your project

Role of Supervisors

- Supervisor = your personal tutor during project
- Helps manage project but does not do work for you
- Provides feedback, advice, and support
- Records attendance at supervision meetings
- Not necessarily an expert in your niche



Supervision Expectations

- Held on campus, attendance mandatory
- Bring progress updates to each meeting
- Group format common, encourages peer learning
- Regular meetings linked to higher success



Managing Workload

- ~600 hours total (~40 hours/week)
- Use PDM Plan to structure tasks and milestones
- Work daily on project with parallel tasks
- Leave buffer time for coding/analysis
- Final report must reflect all work done



Importance of Practical Work

- Coding evidence required for a pass
- Results and analysis must stem from your code
- Viva requires demonstration and explanation
- Start coding early, finish early to allow writing
- Optimisation and experiments should be documented



Purpose of Literature Review

- Provides background and context
- Shows awareness of prior work and methods
- Guides model/method selection
- Supports justification for choices
- Critical element in report assessment



How to Conduct a Literature Review

- Use peer-reviewed papers, not websites or theses
- Search via UH library subscriptions for access
- Select papers based on relevance criteria
- Analyse and critique each paper, not just list
- Highlight gaps or limitations your project addresses



Referencing Guidelines (Harvard)

- All in-text citations must be in reference list
- References listed alphabetically by first author
- Include full details: author, year, title, journal, pages, URL
- Websites require publisher, year, access date
- Fake or missing references = academic offence



Academic Misconduct Defined

- Any unfair advantage in assessment = misconduct
- Includes plagiarism, collusion, contract cheating
- AI-generated text/code without declaration = misconduct
- Fabrication of data or false references is misconduct
- Penalties: fail module or withdrawal from course



Using Turnitin

- All submissions checked via Turnitin
- Similarity score 10–30% is normal
- Copying paragraphs = fail risk
- Focus on originality of expression and analysis
- Reports reviewed by markers, not just % score



Plagiarism, Collusion & AI Tools

- Plagiarism = uncredited use of others' work
- Collusion = shared/copied work between students
- AI tools (e.g. ChatGPT) use must be declared
- Unauthorised AI use = 0 grade, misconduct report
- Code submitted must be your own



Summary of Assessments

- PDM Plan presentation (10%)
- Ethics Test (10%)
- Final Project Report (70%)
- Viva (10%)
- Must complete all to pass

#1 Project and Data Management Plan (PDM)

Project and Data Management Plan

The Project and Data Management (PDM) Plan is a summary of your project topic, the research question, a project plan and a data management plan including an overview of your data.

The PDM plan is assessed by a presentation to your supervisor and supervision group.

Marks are only awarded for the presentation based on the PDM plan.

You must attend the presentation to get any marks.

The plan must be written in either Arial, Times or Calibri font with a font size of 11 and single line spacing.

The whole document must be a maximum of three (3) pages and a maximum of 1000 words.

PDM Plan consists of 3 parts

1. **Project Overview**
2. **Project Plan**
3. **Data Management Plan**

Project Overview

1. **Project Title.**
2. **A short summary of project topic and background**
3. **A Research Question.**
4. **Project Objectives.**
5. **Reference list.**

State Your Project and Objective Clearly

- Present your project title in a complete sentence
- Introduce the context and main goal for your project
- State why this project is important
- Clarify the intended impact for stakeholders



Articulate the Research Question and Specific Project Aims

- Explain the research question that drives your investigation
- Break down project objectives, each as a clear outcome
- Connect objectives to the broader field/problem area
- Indicate any expected deliverables or results

Demonstrate Reference Quality—Include At Least Three Literature in Harvard Format in the PDM

- List at least three peer-reviewed sources in Harvard format
- Comment briefly on the relevance of each reference
- Include active hyperlinks to online sources
- Justify the selection of literature in supporting your research question



Map Out Actionable Steps—Present Your Key Project Tasks and Timeline

- Display a timeline (Gantt chart or schedule) with overlapping tasks and assessment dates
- Bullet point each major task: literature review, data collection, analysis, etc.
- Note the purpose of each individual task
- Highlight planned multitasking to save time and increase project efficiency



Ensure Transparency—Summarize How Data is Collected, Managed, and Validated

- Describe the origin and description of your dataset (country, organization, etc.)
- Specify methods and sources for data collection (website, database, survey, etc.)
- Provide details on data format, anticipated size, and number of entries
- Outline how the validity and relevance are checked

Detail Version Control and Backup Procedures Using GitHub

- Share your GitHub repository link and weekly commitment schedule
- Explain your file naming and versioning system
- State what your ReadMe file will include for others to understand/use your code
- Identify secondary backup location (e.g., OneDrive) and update routine and mention frequency of backup for both data and code.



Protect Data and Guarantee Ethical Use—Describe Security, Permissions, and Compliance

- Describe sharing protocols and how access is managed for staff, markers, and collaborators
- Address GDPR compliance and institutional ethics requirements
- Detail permission status for all data used, and provide evidence if possible
- Demonstrate that data sources were collected ethically by originators



Recap Major Insights and Invite Discussion

- Summarize key project aims and anticipated impact
- Restate unique aspects (innovation, rigor, societal relevance)
- Invite feedback and questions for clarification or deeper understanding

Project Plan

Produce a timeline of tasks and activities that you will be doing to complete the project module.

This can be a chart and/or a list of tasks with dates.

Include the assessments as key dates and allocate time for preparation of the assessments.

Timeline

Simple excel spreadsheet

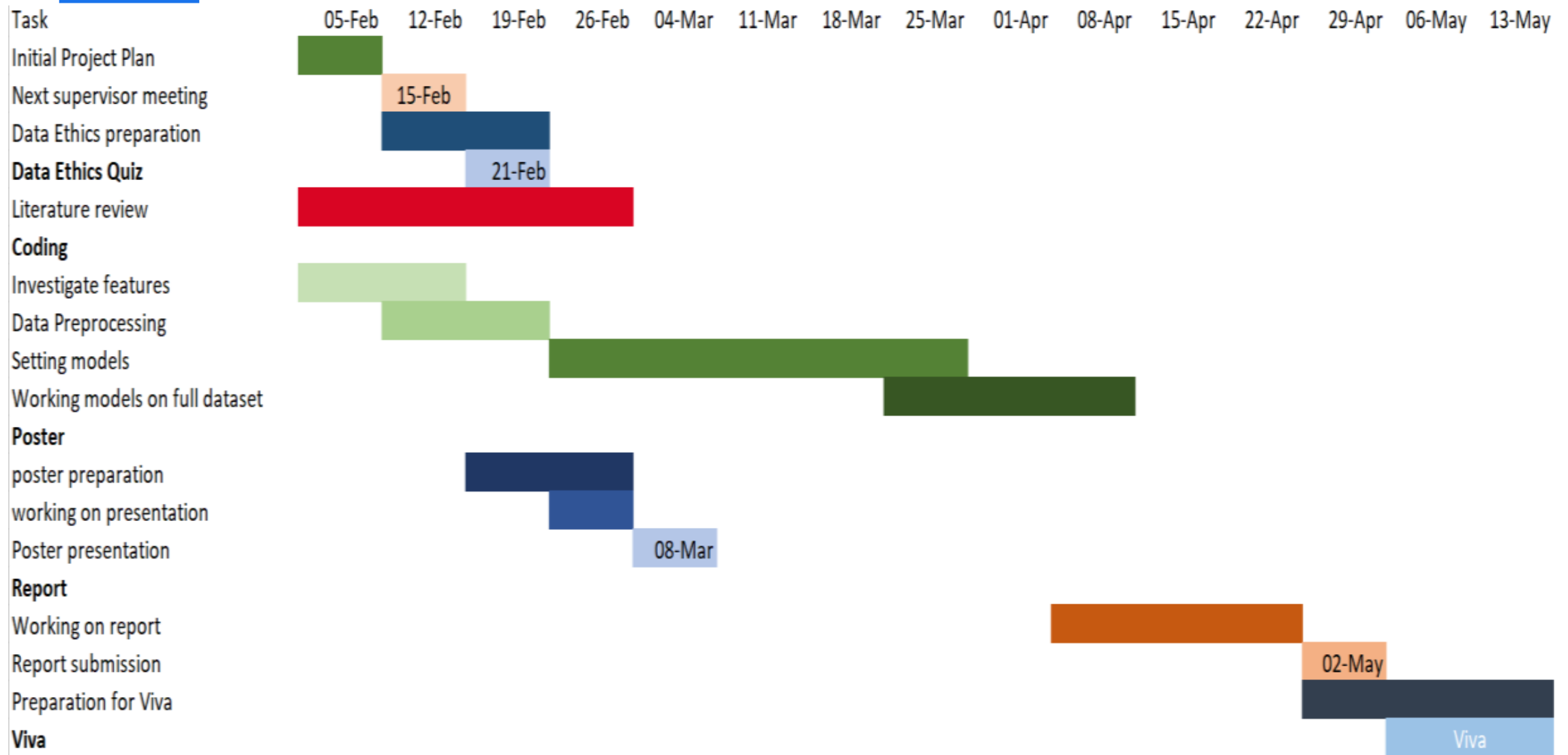
Gantt Chart

Example task list

Task	Description	Start Date	End Date
Literature Review	Conduct a comprehensive review of existing literature on energy demand forecasting.	23/01/2024	23/02/2024
Data Acquisition	Obtain and preprocess the required datasets from the specified sources.	23/01/2024	30/01/2024
Initial Project plan	Prepare and submit IPP	29/01/2024	05/02/2024
Data Analysis	Analyze the datasets using clustering and visualizations.	05/02/2024	19/02/2024
Data ethics Quiz	Prepare and take the quiz	14/02/2024	21/02/2024
Initial Model Development	Develop and train the chosen machine learning models.	30/01/2024	29/02/2024
Comparative Analysis	Compare the performance of the machine learning models.	26/02/2024	11/03/2024
Poster presentation	Preparation of poster and presentation	01/03/2024	11/03/2024
Model Development and Documentation	Build the model and Document the findings and prepare the final report.	12/03/2024	12/04/2024
Final Project report	Prepare and submit final project report	08/04/2024	02/05/2024
Viva preparation	Prepare for Viva	01/05/2024	17/05/2024

Project Timeline

Example Excel timeline



Project Timeline

Example Gantt chart

' DVD '6 FLHQFH '0 VF '3 URNHFV

'2 QJ RLQJ 'L

LLLLL H' VVXSHU'VRUP HMQJ

LLLLL HHS 'GRMV'RQ'Z KDW P 'GRLQJ

'4 LVHUVXUH '5 HYLHZ

LLLLL '6 HOUFK '1 RU- 'UHYLHZ 'UHQYDQVDFQV

'8 & RGLQJ

LLLLL ([SORUVRU 'GDV'DQDQ VLV

LLLLLLL 'QYHWW DM '1 HDVUH

LLLLLLL DVD 'SUH'SURFHMQJ

LLLLL &RGH'DQ RUVKP V

LLLLLLL '6 HMXS'P RGHQ'XVLQ 'VP DQJGDV'V

LLLLLLL '5 XQ'P RGHQ'RQ'XQGDV'VHV

'3 RVMU

LLLLL '3 RVMU'LVWGLD'W

LLLLL '3 RVMU'HGLVQJ

LLLLL '3 RVMU'XSQDQ

'4) LQDQHSRUW

LLLLL 'LVM'XS'EDFN URXQG'VHFVRQ

LLLLL 'LVM'XS'P HMRGRQJ' 'VHFVRQ

LLLLL 'LVM'XS'UHVXQV'VHFVRQ

LLLLL 'LVM'XS'FRQFQVRQV'VHFVRQ

LLLLL &RP SQQMG'LVWGLD'W

LLLLL (GLVGLD'W

LLLLL '6 XEP LVHSRUW

'9 LYD

LLLLL '9 LYD'SUHSUDVRQ

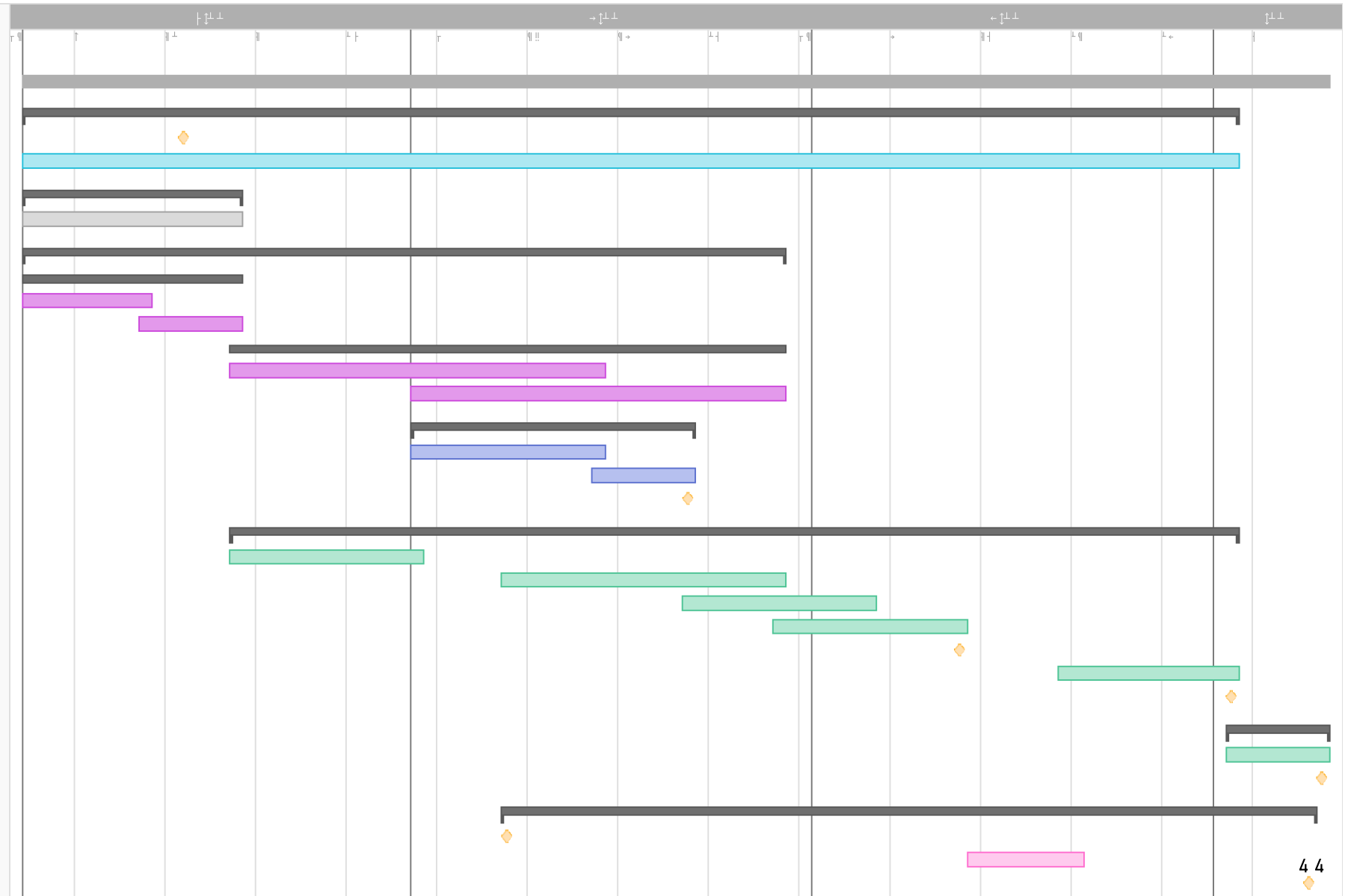
LLLLL '9 LYD'P HMQJ

'3 HUVRQDOFQHQGDU

LLLLL * LV'EUDN'XS '1 RUVXP P HU

LLLLL + RQDQ

LLLLL * LV'EUDN'VR'VFKRRO



Data Management Plan

1. Overview of the dataset

- Background to who, where, when and how data was originally collected/made.

2. Data collection

- Including the specific website address and link to the dataset

3. Metadata

- Including format, how many records, expected size of the data and code files (e.g. Gigabytes)

4. Document control

- Including file naming, version control, use of GitHub and GitHub address.

5. ReadMe file

- A statement of what will be in the ReadMe file.

6. Security and storage

- Including frequency of backups/updates stored and how data will be shared (e.g. with staff). Where will the data be stored?

7. Ethical requirements



Marking Criteria PDM Plan



PDM Plan (60%)

- Quality of Project Overview
- Quality of Project Plan
- Quality of Data Management Plan

Presentation (40%)

- Quality of presentation
- Response to questions

Some things to consider when planning

Prepare a good research question that is answered by your project.

You must do at least one model you develop yourself. You cannot use all pre-trained models. If you use a pre-trained model, then you must know what layers/structure is used in the model and what data it was trained on. You should not blindly use it you must understand it.

It is better to do one model well than have 5 done poorly. “well” means you have developed the model yourselves (e.g. the layers of a neural net) and you have optimised the results and tried different parameters and hyperparameters (this is part of the research) to get the best outputs you can.

If you are in a supervision group with multiple projects the same, then make sure you have a different research aim and/or dataset. Ideally both. You should use different sets of models (some can be the same but not all of them).

Some things to consider when planning

If you are planning to use Kaggle then try to find other data instead that is not from Kaggle. Having data that you extract and pre-process yourself is better (you demonstrate more skills).

Combining 2 datasets together is better, then you are demonstrating more complex pre-processing

You can generate your own dataset e.g. using an autoencoder to generate images.

You MUST NOT collect data through a survey including online, phone, asking friends (you need UH ethical approval to do this).

You cannot extract information from social media sites. This is people's personal information and you need UH ethical approval to do this. If you use this type of information from Kaggle or other websites you have to check carefully that you have permission to use it.



PDM Plan Assessment

- 60%: project overview, plan, data, ethics
- 40%: presentation & Q&A
- Presentation (10 mins) + Q&A (5-15mins)
- Must attend in person (otherwise 0%)
- Practice for clarity and timing

#2 Ethics Test

Ethics Test

- On campus, multiple choice
- Covers ethics, law, referencing, misconduct
- Based on lectures and reading list
- Attendance mandatory on set date
- No extensions or late attempts

#3 Final Project Report



Final Project Report

- Worth 70%, max 7,500 words
- Includes research question, methods, results, conclusions
- Professional format required (Arial/Times/Calibri)
- Include dataset, ethics, EDA, analysis
- Code uploaded on GitHub with ReadMe file

#4 Viva



Viva Assessment

- Technical interview with supervisor + second marker
- 20–30 minutes, on campus
- Demonstrate code, results, and understanding
- Answer questions clearly and confidently
- Passing viva = mandatory to pass module



Key Success Factors

- Start early with coding and literature review
- Manage time with clear weekly plan
- Meet supervisor regularly and prepare
- Ensure dataset is ethical and appropriate
- Backup work in multiple locations



Final Advice

- Treat project as professional research experience
- Be honest, thorough, and critical
- Ask for help early when problems arise
- Practise viva answers (esp. about code)
- Aim to produce a report for employers

Thank You

ANY QUESTIONS?

